Experiment 16

Experiment to understand EDA-Quantitative and Qualitative analysis

Aim:

Understand the importance of EDA-Quantitative and Qualitative analysis.

Code:

```
import seaborn as sns
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
%matplotlib inline

tips = sns.load_dataset('tips')
tips.head()

sns.displot(tips.total_bill, kde=True)
sns.displot(tips.total_bill, kde=False)

sns.jointplot(x=tips.tip, y=tips.total_bill)
sns.jointplot(x=tips.tip, y=tips.total_bill, kind="reg")
sns.jointplot(x=tips.tip, y=tips.total_bill, kind="hex")

sns.pairplot(tips)

tips.time.value_counts()
sns.pairplot(tips, hue='time')
sns.pairplot(tips, hue='day')

sns.heatmap(tips.corr(numeric_only=True), annot=True)

sns.boxplot(tips.total_bill)
sns.boxplot(tips.tip)
```

```
sns.countplot(tips.day)
sns.countplot(tips.sex)

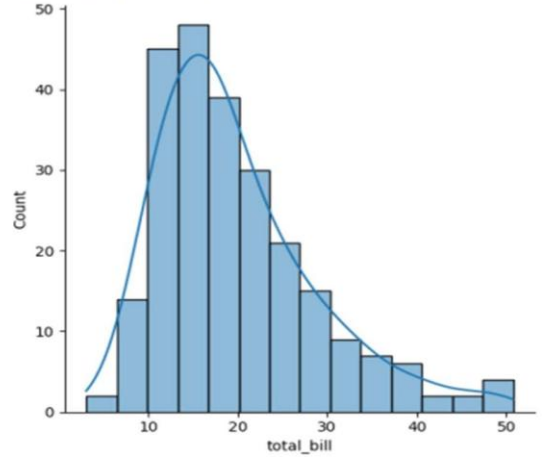
tips.sex.value_counts().plot(kind='pie')
tips.sex.value_counts().plot(kind='bar')

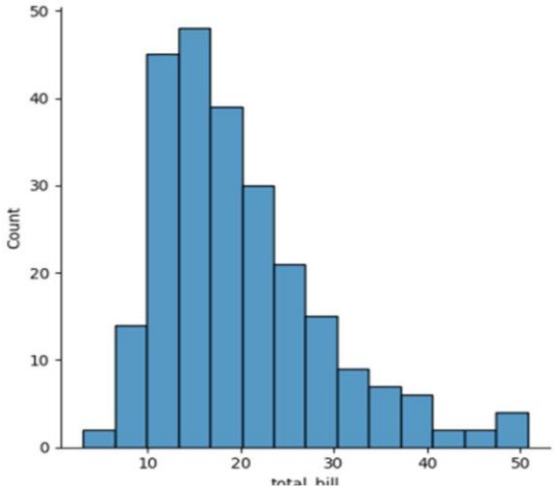
sns.countplot(tips[tips.time=='Dinner']['day'])
```

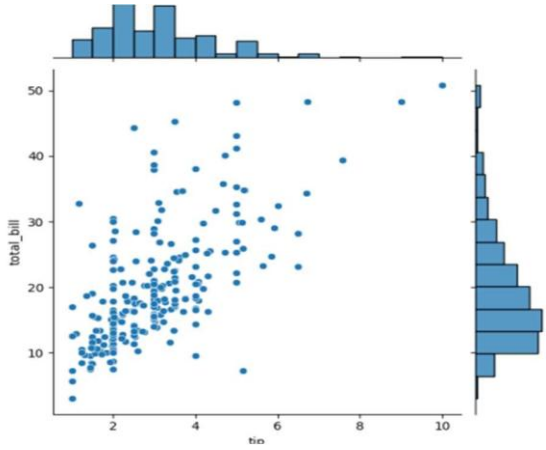
Output:

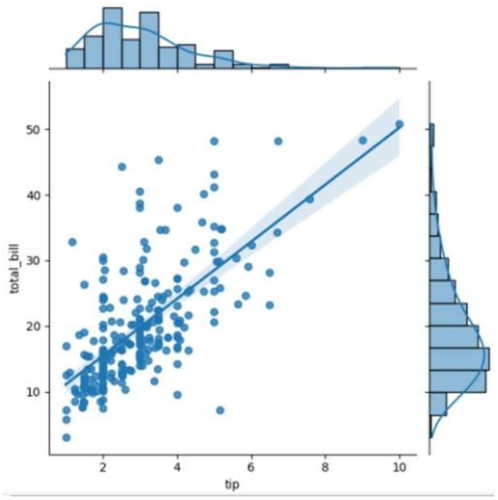
	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4

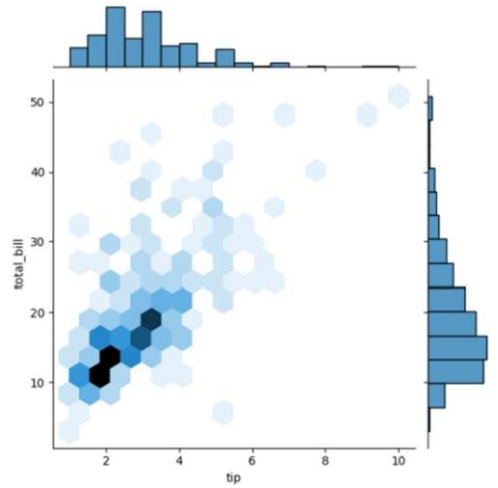
```
<seaborn.axisgrid.FacetGrid at 0x79bb4c7ea680>
```

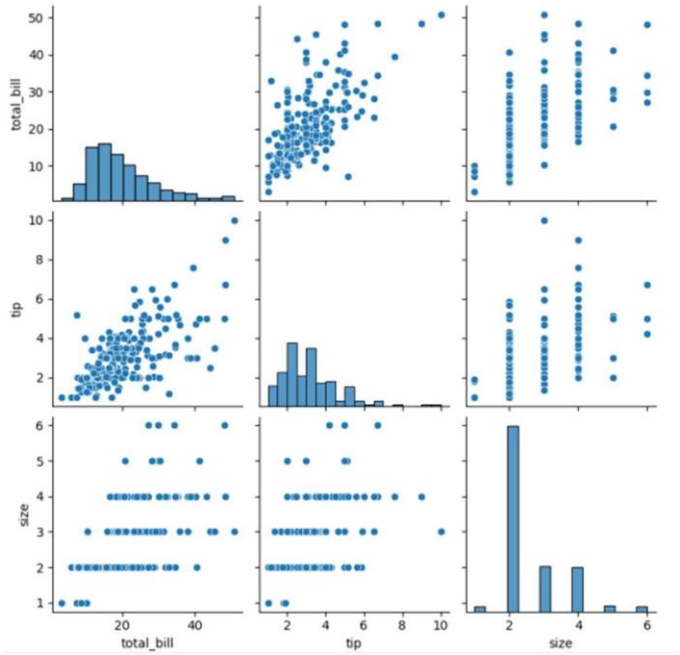




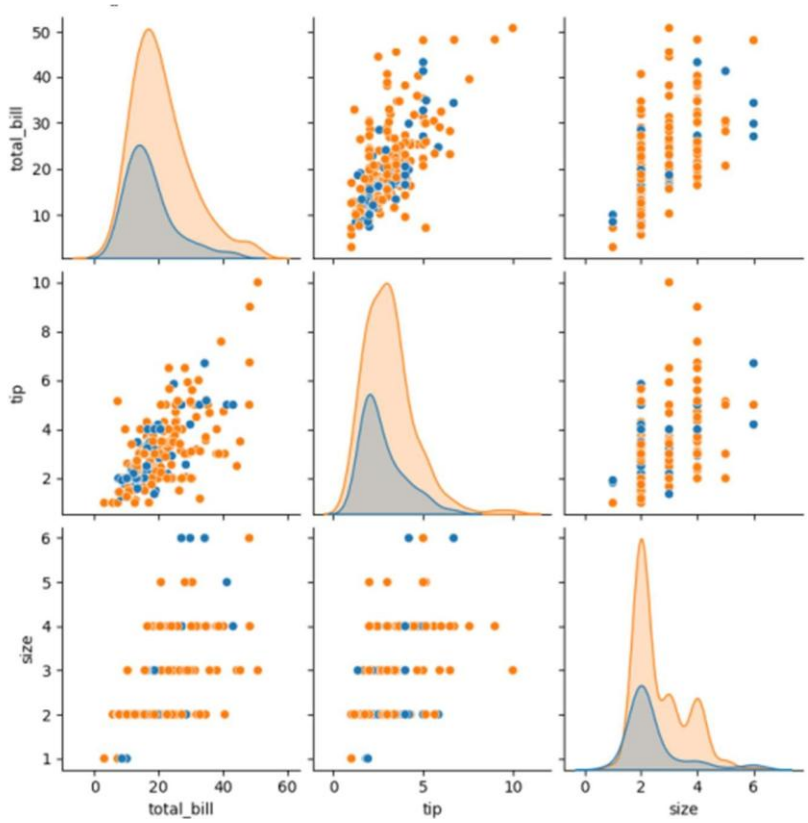



total_bill
tip


total_bill
tip


total_bill
tip
size

time	count
Dinner	176
Lunch	68



Result:

EDA on the Seaborn 'tips' dataset successfully revealed both quantitative and qualitative insights. Distribution plots showed right-skewed total bill amounts, while categorical plots indicated that dinner tips are higher than lunch, and male customers tip slightly more on average.